

# ECE 447

## Lesson 34

## GR2 Review Day



UNITED STATES  
AIR FORCE  
ACADEMY

# SCHEDULE AND ADMIN

- **Schedule.**
  - Lesson 34 - **GR2 Review Day** (today)
  - Lesson 35 - GR2 (Tuesday, 17 Nov)
  - Lessons 36-38 - Binary Digital System Performance, Error Correction
  - Lesson 39 - SDR Lab 8: Capstone image transfer
  - Lesson 40 - Advanced topics: OFDM, MIMO, CDMA
  - Lesson 41 - Course review
- Admin
  - **HW7.** Due tonight (Lesson 34, 13 Nov) to Gradescope.

# CHP 5 REVIEW: LESSONS 18-20, HWs 3 & 4

## • Sampling

- Impulse train in time  $\leftrightarrow$  impulse train in frequency
- Creates copies of message spectrum in frequency
- Sinc is ideal interpolator, if  $f_s \geq 2B$
- Non-ideal pulses have "filtering" effect on sampled signal - equalizers reverse the filtering to obtain original signal
- What two requirements do we have for *distortionless* filtering?
- $G(f)e^{-j2\pi ft_0} = E(f)\tilde{G}(f) = E(f)P(f)\frac{1}{T_s} \sum_n G(f - nf_s)$ , so make equalizer transfer function

$$E(f) = \begin{cases} T_s e^{-j2\pi ft_0} \cdot |P(f)|^{-1}, & |f| \leq B \\ \text{Flexible}, & B < |f| < f_s - B \\ 0, & |f| \geq f_s - B \end{cases}$$

- Just need to invert  $P(f)$  and plug into  $E(f)$ ! Example: 5.1-5a - find  $E(f)$  if  $p(t) = \Pi(\frac{t}{T_s} - \frac{1}{2})$

# CHP 5 REVIEW: LESSONS 18-20, HWs 3 & 4

- PCM

- Just sample, quantize, and encode
- Message peak at  $\pm m_p$ , so each bin is  $2m_p/L$ , where  $L = 2^n$  levels,  $n$  = number of bits
- Each additional bit improves the SNR how much?
- Max data rate per bandwidth is  $\frac{2\text{bps}}{\text{Hz}}$  - assuming a noiseless channel w/ bandwidth  $B$
- **If sampling exactly at Nyquist**,  $f_s = 2B$ , then min req'd channel bandwidth is  $B_T = nB$
- **If sampling at greater than Nyquist**,  $f_s > 2B$ , then min req'd channel bandwidth is  $B_T = \frac{nf_s}{2}$  or simply the data rate  $nf_s$  divided by 2
- Be able to solve problems like 5.2-3 or Example 5.2, where a max quantization error is given as a percentage of  $m_p$  and  $f_s$  is given as a percent of the Nyquist rate

# CHP 6 REVIEW: LESSONS 22-27, HWs 5 & 6

- **Digital Communications**

- **Line codes**
- Be able to plot  $y(t) = \sum_k a_k p(t - kT_b)$ , i.e., the baseband waveform
- Be able to use generic PSD equation in (6.11c)? Maybe, but I'd provide values for  $R_n$
- Band-limited vs. time-limited
- **ISI**
- Nyquist's First Criterion for Zero ISI:

$$p(t) = \begin{cases} 1, & t = 0 \\ 0, & t = \pm nT_b \end{cases}$$

- Excess bandwidth  $f_x$ , roll-off factor  $r$
- GR will **clearly** use  $R_b$  for bit rate and  $R_s$  for symbol rate/ baud
- How do we accomplish timing synchronization?
- Parts of eye diagrams

# CHP 7 REVIEW: LESSONS 30-31, HW 7

- **Probability**

- Bernoulli trials using bits
- Conditional/total probability
- Gaussian RV, Q function
- Expected value, mean square, and variance
- Central limit theorem